Lab 11 - Grading the professor, Pt. 2

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Introduction

In this lab we revisit the professor evaluations data we modeled in Lab 10. In the last lab we modeled evaluation scores using a single predictor at a time. This time we will use multiple predictors to model evaluation scores.

This lab builds directly on Lab 10. Make sure you understand single predictor models before working with multiple predictors.

For full context, review Lab 10's introduction before continuing.

Estimated time: 60-90 minutes

Learning Goals

By the end of this lab, you will be able to:

- Fit linear regression models with multiple predictors
- Interpret regression coefficients in multiple regression
- Compare simple and multiple regression models
- Understand adjusted R^2 and when to use it
- Interpret interaction effects between predictors
- Make predictions using multiple regression models

Prerequisites

Before you begin, make sure you have:

- **Completed Lab 10** (this is essential!)
- Watched lectures on multiple regression
- Understand single predictor regression
- Understand the concepts of R^2 and adjusted R^2
- Forked the lab-instructions repository

Getting Started

Navigate to your forked `lab-instructions` repository in JupyterHub, open the `Lab11` folder, and open the R Markdown document `lab-11.Rmd`.

Verify Your Setup

Before proceeding:

Step 1. Open `lab-11.Rmd` in RStudio

Step 2. Check that the Git pane shows YOUR username (not the course organization)

Step 3. Click **Knit** to make sure the document compiles

Warm Up

Step 1: Update the YAML, changing the author name to your name.

Step 2: Knit the document.

Step 3: Commit your changes with message "Updated author name".

Step 4: Push to GitHub and verify the changes are visible in your repo.

Knit, commit with message "Completed warm up", and push your changes.

Packages

We'll use the **tidyverse** package for much of the data wrangling and visualization, the **tidymodels** package for modeling and inference, and the data lives in the **openintro** package. These packages are already installed for you. You can load them by running the following in your Console:

```
library(tidyverse)
library(tidymodels)
library(openintro)
```

Data

The data can be found in the **openintro** package, and it's called `evals`. Since the dataset is distributed with the package, we don't need to load it separately; it becomes available to us when we load the package. You can find out more about the dataset by inspecting its documentation, which you can access by running `?evals` in the Console or using the Help menu in RStudio to search for `evals`. You can also find this information here.

Exercises

Exercise 1

Load the data by including the appropriate code in your R Markdown file.

Note: Since `evals` comes from the openintro package, it's automatically available when you load the package. However, it's good practice to verify the data loaded correctly.

```
# The data is already available, but let's verify it loaded
glimpse(evals)
```

```
## Rows: 463
## Columns: 23
## $ course_id     <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1~
## $ prof_id       <int> 1, 1, 1, 1, 2, 2, 2, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5,~
## $ score         <dbl> 4.7, 4.1, 3.9, 4.8, 4.6, 4.3, 2.8, 4.1, 3.4, 4.5, 3.8, 4~
## $ rank          <fct> tenure track, tenure track, tenure track, tenure track, ~
## $ ethnicity     <fct> minority, minority, minority, minority, not minority, no~
## $ gender        <fct> female, female, female, female, male, male, male, male, ~
## $ language      <fct> english, english, english, english, english, english, en~
## $ age           <int> 36, 36, 36, 36, 59, 59, 59, 51, 51, 40, 40, 40, 40, 40, ~
## $ cls_perc_eval <dbl> 55.81395, 68.80000, 60.80000, 62.60163, 85.00000, 87.500~
## $ cls_did_eval  <int> 24, 86, 76, 77, 17, 35, 39, 55, 111, 40, 24, 24, 17, 14,~
## $ cls_students  <int> 43, 125, 125, 123, 20, 40, 44, 55, 195, 46, 27, 25, 20, ~
## $ cls_level     <fct> upper, upper, upper, upper, upper, upper, upper, upper, ~
## $ cls_profs     <fct> single, single, single, single, multiple, multiple, mult~
## $ cls_credits   <fct> multi credit, multi credit, multi credit, multi credit, ~
## $ bty_f1lower   <int> 5, 5, 5, 5, 4, 4, 4, 5, 5, 2, 2, 2, 2, 2, 2, 2, 2, 7, 7,~
## $ bty_f1upper   <int> 7, 7, 7, 7, 4, 4, 4, 2, 2, 5, 5, 5, 5, 5, 5, 5, 5, 9, 9,~
## $ bty_f2upper   <int> 6, 6, 6, 6, 2, 2, 2, 5, 5, 4, 4, 4, 4, 4, 4, 4, 4, 9, 9,~
## $ bty_m1lower   <int> 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 7, 7,~
## $ bty_m1upper   <int> 4, 4, 4, 4, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 6, 6,~
## $ bty_m2upper   <int> 6, 6, 6, 6, 3, 3, 3, 3, 3, 2, 2, 2, 2, 2, 2, 2, 2, 6, 6,~
## $ bty_avg       <dbl> 5.000, 5.000, 5.000, 5.000, 3.000, 3.000, 3.000, 3.333, ~
## $ pic_outfit    <fct> not formal, not formal, not formal, not formal, not form~
## $ pic_color     <fct> color, color, color, color, color, color, color, color, ~
```

How many observations are in the dataset? Use inline code to report.

Your answer: The dataset has 463 observations.

The dataset has 23 variables.

Checkpoint: You should see 463 observations and 23 variables.

Part 1: Simple Linear Regression (Review)

Exercise 2

Fit a linear model (one you have fit in Lab 10): `score_bty_fit`, predicting average professor evaluation `score` based on average beauty rating (`bty_avg`) only. Write the linear model, and note the R^2 and the adjusted R^2.

```
# Remember to change eval=FALSE to eval=TRUE!
score_bty_fit <- linear_reg() %>%
  set_engine("lm") %>%
  fit(score ~ bty_avg, data = evals)

# View coefficients
tidy(score_bty_fit)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)   3.88      0.0761     51.0  1.56e-191
## 2 bty_avg       0.0666    0.0163      4.09 5.08e-  5
```

```
# View model statistics including R²
glance(score_bty_fit)
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic   p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>     <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1    0.0350        0.0329 0.535      16.7 0.0000508     1  -366.  738.  751.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Write the linear model equation:

$$\widehat{score} = 3.88 + 0.0666 \times bty\_avg$$

R^2 value: 0.0350
Adjusted R^2 value: 0.0329

Knit, commit with message "Completed Exercises 1-2", and push your changes.

Part 2: Multiple Linear Regression

Exercise 3

Fit a linear model (one you have fit in Lab 10): `score_bty_gen_fit`, predicting average professor evaluation `score` based on average beauty rating (`bty_avg`) and `gender`. Write the linear model, and note the R^2 and the adjusted R^2.

```
# Remember to change eval=FALSE to eval=TRUE!
score_bty_gen_fit <- linear_reg() %>%
```

```
  set_engine("lm") %>%
  fit(score ~ bty_avg + gender, data = evals)

# View coefficients
tidy(score_bty_gen_fit)
```

```
## # A tibble: 3 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)   3.75      0.0847     44.3  6.23e-168
## 2 bty_avg       0.0742    0.0163      4.56 6.48e-  6
## 3 gendermale    0.172     0.0502      3.43 6.52e-  4
```

```
# View model statistics
glance(score_bty_gen_fit)
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic     p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>       <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1    0.0591        0.0550 0.529      14.5 0.000000818     2  -360.  729.  745.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Write the linear model equation:

$$\widehat{score} = 3.75 + 0.0742 \times bty\_avg + 0.172 \times gendermale$$

R^2 value: 0.0591
Adjusted R^2 value: 0.0550

Exercise 4

Interpret the slopes and intercept of `score_bty_gen_fit` in context of the data.

Interpret the intercept:

The intercept represents that when the average beauty rating of the professor is 0 stars and the professor is female, then their average evaluation score is expected to be 3.75 stars.

Interpret the slope for bty_avg:

Holding gender constant, for every one star increase in beauty rating, we expect there to be a 0.0742 star increase in professor evaluation score.

Interpret the slope for gendermale:

Holding beauty rating constant, male professors are expected to score 0.172 stars higher in average evaluation score than female professors.

Important: In multiple regression, we interpret each coefficient "holding all other variables constant."

Exercise 5

What percent of the variability in `score` is explained by the model `score_bty_gen_fit`?

Use the R^2 value from Exercise 3.

Approximately 5.91% of the variability in professor evaluation scores is explained by beauty rating and gender.

Exercise 6

What is the equation of the line corresponding to *just* male professors?

Hint: For male professors, gendermale = 1.

Starting with: $\widehat{score} = b_0 + b_1 \times bty\_avg + b_2 \times gendermale$

For males (gendermale = 1):

$$\widehat{score} = 3.75 + 0.0742 \times bty\_avg + 0.172 \times 1$$

Simplified:

$$\widehat{score} = (b_0 + b_2) + b_1 \times bty\_avg$$

With your actual values:

$$\widehat{score} = 3.922 + 0.0742 \times bty\_avg$$

Knit, commit with message "Completed Exercises 3-6", and push your changes.

Exercise 7

For two professors who received the same beauty rating, which gender tends to have the higher course evaluation score?

Look at the coefficient for gendermale. Is it positive or negative?

Your answer:

If the coefficient is positive: Male professors tend to have higher scores. If the coefficient is negative: Female professors tend to have higher scores.

The coefficient is: positive

Therefore: If a professor is male, I expect the average professor evaluation rating to be 0.172 stars higher than if the professor was female, with all else held constant.

By how much? 0.172 stars.

Exercise 8

How does the relationship between beauty and evaluation score vary between male and female professors?

Look at the coefficient for bty_avg in the multiple regression model.

The coefficient for bty_avg is 0.0742.

Your answer:

The relationship between beauty and evaluation score does not vary between male and female professors because the model is not an interaction effects model, and gender does not impact the beauty coefficient.

Hint: In this model, both genders have the same slope for bty_avg. The lines are parallel!

Exercise 9

How do the adjusted R^2 values of `score_bty_gen_fit` and `score_bty_fit` compare? What does this tell us about how useful `gender` is in explaining the variability in evaluation scores when we already have information on the beauty score of the professor?

Compare the adjusted R^2 values:

score_bty_fit adjusted R^2: 0.0329 (from Exercise 2)

score_bty_gen_fit adjusted R^2: 0.0550 (from Exercise 3)

Difference: 0.0221

Interpretation:

The adjusted R^2 increased by 0.0221 when we added gender to the model. This tells us that gender explains 2.21% of the variability in professor evaluation scores not described by beauty.

Note: Adjusted R^2 accounts for the number of predictors, so it's better for comparing models with different numbers of predictors.

Exercise 10

Compare the slopes of `bty_avg` under the two models (`score_bty_fit` and `score_bty_gen_fit`). Has the addition of `gender` to the model changed the parameter estimate (slope) for `bty_avg`?

Slope of bty_avg in score_bty_fit: 0.0666 (from Exercise 2)

Slope of bty_avg in score_bty_gen_fit: 0.0742 (from Exercise 3)

Has it changed?

The bty_avg slope has changed. The difference is 0.0076.

Why might this happen?

If male professors have a higher average beauty rating than female professors or vice versa, then adding gender as a predictor variable could change the weight of bty_avg in the model.

Hint: If beauty and gender are related (e.g., if male and female professors have different average beauty ratings), adding gender can change the bty_avg coefficient.

Knit, commit with message "Completed Exercises 7-10", and push your changes.

Exercise 11

Create a new model called `score_bty_rank_fit` with `gender` removed and `rank` added in. Write the equation of the linear model and interpret the slopes and intercept in context of the data.

```
# Remember to change eval=FALSE to eval=TRUE!
score_bty_rank_fit <- linear_reg() %>%
  set_engine("lm") %>%
  fit(score ~ bty_avg + rank, data = evals)

# View coefficients
tidy(score_bty_rank_fit)

## # A tibble: 4 x 5
##   term             estimate std.error statistic   p.value
##   <chr>               <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)        3.98      0.0908     43.9  2.92e-166
## 2 bty_avg            0.0678    0.0165      4.10 4.92e-  5
## 3 ranktenure track  -0.161     0.0740     -2.17 3.03e-  2
## 4 ranktenured       -0.126     0.0627     -2.01 4.45e-  2

# View model statistics
glance(score_bty_rank_fit)

## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic   p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>     <dbl> <dbl>  <dbl> <dbl> <dbl>
```

```
## 1    0.0465        0.0403 0.533      7.46 0.0000688     3  -363.  737.  758.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

Note: `rank` has three levels: teaching (reference), tenure track, and tenured.

Write the linear model equation:

$$\widehat{score} = 3.98 + 0.067 \times bty\_avg - 0.161 \times ranktenuretrack - 0.126 \times ranktenured$$

Interpret the intercept:

For teaching faculty with a beauty rating of 0, the expected evaluation score is 3.98 stars.

Interpret the slope for bty_avg:

Holding rank constant, for every one unit increase in beauty rating, the average professor evaluation score should increase by 0.0678 stars.

Interpret the slope for ranktenuretrack:

Holding beauty rating constant, tenure track faculty are expected to score 0.161 stars lower than teaching faculty.

Interpret the slope for ranktenured:

Holding beauty rating constant, tenured faculty are expected to score 0.126 stars lower than teaching faculty.

Adjusted R^2 for this model: 0.0403

How does this compare to score_bty_gen_fit? This R^2 value is less than the adjusted R^2 value of score_bty_gen_fit of 0.0550. This means that the score_bty_gen_fit described 1.47% more of the variance in mean professor evaluation scores.

Knit, commit with message "Completed Exercise 11", and push your changes.

Common Errors and Troubleshooting

Multiple regression errors:

- **Coefficients look wrong** → Check that you included the `+` between predictors
- **Can't interpret "holding other variables constant"** → This means we compare observations that have the same value for other predictors
- **Confused about which R^2 to use** → Use adjusted R^2 when comparing models with different numbers of predictors

Model comparison errors:

- **Can't see difference in R^2** → Look at more decimal places in `glance()` output

- **Slopes changed a lot** → This can happen if predictors are correlated
- **Not sure which model is better** → Higher adjusted R^2 generally means better, but also consider context and interpretability

Interpretation errors:

- **Forgot "holding other variables constant"** → This is crucial in multiple regression!
- **Confused about categorical variables** → Remember the reference level; coefficients are relative to that
- **Intercept doesn't make sense** → That's okay if it represents values outside the data range (like beauty = 0)

General issues:

- **Code chunk not running** → Make sure you changed `eval=FALSE` to `eval=TRUE`
- **Can't remember coefficients** → Run `tidy(model_name)` again
- **Need more decimal places** → Use `tidy(model_name) %>% print(n = Inf, width = Inf)`

Reflection

Key differences between simple and multiple regression:

- **Simple regression:** One predictor, one slope to interpret
- **Multiple regression:** Multiple predictors, interpret each "holding others constant"
- **R^2 vs. Adjusted R^2:** Adjusted R^2 penalizes adding unnecessary predictors

Important concepts:

- Coefficients in multiple regression show the effect of one predictor while controlling for others
- Adding predictors can change existing coefficients if predictors are correlated
- Adjusted R^2 is better for model comparison than regular R^2
- Higher adjusted R^2 doesn't always mean a "better" model - consider interpretability and context

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To submit to Canvas:

Step 1. In RStudio, click the **Knit** dropdown menu (next to the Knit button)

Step 2. Select **Knit to tufte_handout** to generate a PDF

Step 3. Download the PDF file from the Files pane

Step 4. Upload the PDF to Canvas

Final Checkpoint: Visit your GitHub repo one more time to confirm all your work is there. We will grade what we see in your repo on GitHub!